

Mama GCS

Hardware information request

Send this form to the vehicle hardware, firmware and safety teams. Return one completed copy with the requested evidence attachments.

Why this is needed

Mama GCS currently receives ArduPilot MAVLink telemetry and displays a map. Its VESC work is read-only groundwork; live VESC input, movement, spraying, hydraulics and app emergency-stop commands are **not operational**. The information below is needed to implement each connection safely and accurately.

How to answer

For each ID, enter the exact value or an attachment filename/page reference. If unknown, write **UNKNOWN** and who will verify it; if absent, write **NOT INSTALLED**. Do not use a guessed model, channel, limit or safe state.

Priority labels: **NOW** = needed for current read-only integration; **CONTROL** = required before any vehicle/implement command can be considered; **FEATURE** = needed for a later requested capability.

Security and safety boundary

Do not send passwords, private keys, API keys or full network credentials. Provide configuration method and trust policy instead. A completed form does not authorize operation: bench and field validation, independent physical safety systems and explicit sign-off remain mandatory.

Response cover sheet

Vehicle / build ID _____

Hardware lead _____

Firmware lead _____

Safety reviewer _____

Response date / revision _____

Return package: this completed form; as-built diagrams; model/firmware inventory; ArduPilot parameter export; redacted telemetry samples; hardware/protocol documentation; and safety-test records where available.

A Vehicle architecture and MAVLink link

Return the as-built vehicle topology, not a proposed design. Label every cable, bus, gateway and power domain.

A1 NOW	Vehicle identity UGV make/model, variant, vehicle serial/asset ID, drive layout and installed payloads. State whether this is a prototype or production build.	Response / attachment ref:
A2 NOW	As-built block diagram Show autopilot, companion computer, VESCs, radios, GNSS, battery/BMS, spray, hydraulics and Android device; label physical interfaces and power sources.	Response / attachment ref:
A3 NOW	Autopilot details Flight-controller model/board ID; ArduPilot Rover firmware version; configured vehicle type; parameter export and date captured.	Response / attachment ref:
A4 NOW	MAVLink configuration Dialect/version, system/component IDs, serial/UDP ports, baud rates, routing, signing policy, and observed messages/rates. Supply a short redacted telemetry log.	Response / attachment ref:
A5 NOW	Phone link path Exact data path from autopilot to phone: Wi-Fi/USB/Bluetooth/radio, gateway, IP/UDP ports, unicast vs broadcast, and which side initiates each connection.	Response / attachment ref:
A6 NOW	Gateway/companion computer Model, OS/firmware, installed routing software/version, interfaces to autopilot and VESCs, and ownership of telemetry forwarding.	Response / attachment ref:
A7 CONTROL	Command path and link policy Proposed return path for commands, authority owner, reconnect/background expectations, and what each device does when the phone or link disappears.	Response / attachment ref:

B VESC controllers and drivetrain

These facts unblock a source-specific, read-only VESC adapter. Provide one row or attachment per controller if models differ.

B1 NOW	Controller inventory Controller count; exact model, hardware revision and firmware version for each; stable CAN ID/UART address and physical motor position.	Response / attachment ref:
B2 NOW	Electrical topology As-wired CAN/UART/gateway diagram, bus bitrate or baud rate, termination, isolation and connector pinout. Identify the device that reads each VESC.	Response / attachment ref:
B3 NOW	Telemetry route and format Exact route from VESC to Android; protocol/message IDs, byte layout and documented units, or gateway API/dialect. Include captured raw and decoded sample frames.	Response / attachment ref:
B4 NOW	Available measurements For each controller: motor and controller temperature, mechanical RPM, electrical RPM, motor/input current, input voltage, duty ratio and fault code; mark unavailable fields.	Response / attachment ref:
B5 NOW	Rate and identity Actual update frequency per field/controller, timestamp source, missing-value encoding, controller-disconnect indication and maximum acceptable sample age.	Response / attachment ref:
B6 CONTROL	Motor and wheel data Motor model, pole pairs, gear ratio, wheel diameter/circumference, mounting side and sign convention; document how forward/reverse and road speed are derived.	Response / attachment ref:
B7 CONTROL	Limits, faults and braking Manufacturer-rated voltage/current/temperature, configured VESC limits, fault-code definitions, reset process, regenerative/mechanical braking and safe state on controller loss.	Response / attachment ref:

C Physical safety and motion authority

A telemetry connection is not evidence that movement is safe. Attach test results signed by the responsible safety engineer.

C1 CONTROL	Independent emergency stop Switch/relay/contactor models, wiring diagram, which energy paths are interrupted, measured stop behavior and manual reset procedure.	Response / attachment ref:
C2 CONTROL	Deadman and operator presence Physical or vehicle-side mechanism, owner, expiry timeout, loss behavior and evidence that it cannot be bypassed by app UI state.	Response / attachment ref:
C3 CONTROL	Link-loss failsafe Timeouts and actions in autopilot, gateway, motor controllers, spray and hydraulics for lost phone, MAVLink, RC or power. Include bench-test evidence.	Response / attachment ref:
C4 CONTROL	Authority arbitration Who can control the UGV (RC, onboard autonomy, Android GCS, service tool), priority order, handover rules and manual override behavior.	Response / attachment ref:
C5 CONTROL	Mode/arming policy Allowed Rover modes and arming conditions for each action; supported arm/disarm and stop semantics; required sensor and interlock states.	Response / attachment ref:
C6 CONTROL	Motion command mapping Validated throttle/steering or velocity command protocol, units and bounds, ACK/result feedback, physical-state confirmation and command timeout behavior.	Response / attachment ref:
C7 CONTROL	Acceptance test plan Named test owner, controlled test site, restrained/wheels-off-ground first test, pass/fail criteria, rollback and incident response procedure.	Response / attachment ref:

D Power, positioning and vehicle sensors

Identify the actual sensor source for each reading. Provide rated limits separately from software alert thresholds.

D1 NOW	Battery architecture Pack count, chemistry, nominal/max/min voltage, capacity, BMS model, series/parallel topology and which pack powers each subsystem.	Response / attachment ref:
D2 NOW	Battery telemetry Voltage/current/temperature/remaining-charge sensors, source protocol and message IDs, units, calibration and pack ID mapping; mark unsupported fields.	Response / attachment ref:
D3 CONTROL	Power protection Fuse/contactor/precharge arrangement, undervoltage/overcurrent/thermal behavior, safe shutdown and validated operating thresholds.	Response / attachment ref:
D4 NOW	GNSS and RTK Receiver/base/correction-source models, firmware, antenna placement, correction path, coordinate datum and expected fix status/accuracy.	Response / attachment ref:
D5 NOW	Motion sensors IMU, compass, wheel encoder and odometry hardware; mounting orientation, calibration status, data route and expected update rates.	Response / attachment ref:
D6 FEATURE	Other sensing Cameras, obstacle sensors, proximity/bumper switches, environmental or implement sensors required by this vehicle; models and data interfaces.	Response / attachment ref:
D7 FEATURE	Health limits Approved machine-specific warning/critical limits and expected telemetry rates for battery, GPS, motors and sensors; name the approving engineer.	Response / attachment ref:

E Spray equipment and agricultural payload

Do not infer that a command means spraying has started or stopped. Identify feedback and physical off-state evidence.

E1 FEATURE	Spray inventory Pump model(s), tank, regulator, boom sections, nozzle count/type and stable nozzle/section IDs; attach plumbing and electrical diagrams.	Response / attachment ref:
E2 FEATURE	Actuator mapping For each pump/nozzle/valve: controller, relay/servo/CAN channel, polarity, normally-open/closed behavior, command units and safe default.	Response / attachment ref:
E3 FEATURE	Flow and pressure Sensor models, installation point, range, units, calibration curve, update rate, fault encoding and telemetry route.	Response / attachment ref:
E4 CONTROL	Spray interlocks Required speed, location, pressure, tank level, mode and operator conditions; dry-run/priming and geofence behavior.	Response / attachment ref:
E5 CONTROL	Stop confirmation Independent feedback proving pump off and valves closed; timeout, leak/drip behavior and actions on link or power loss.	Response / attachment ref:
E6 FEATURE	Application targets Required flow/rate units, target-rate range, section control behavior, calibration procedure and chemical-handling constraints.	Response / attachment ref:
E7 FEATURE	Fault handling Pump/nozzle/sensor fault codes, detection source, manual recovery and which conditions prohibit operation.	Response / attachment ref:

F Hydraulics and auxiliary actuators

Provide the physical neutral/de-energized behavior of every hydraulic output, not only the commanded state.

F1 FEATURE	Hydraulic inventory Pump, controller, manifold, valves and cylinders/implements; exact models, ratings, firmware and as-built hydraulic schematic.	Response / attachment ref:
F2 FEATURE	Channel mapping For each pump/valve: controller and output address, command encoding, direction/polarity, neutral position and de-energized state.	Response / attachment ref:
F3 FEATURE	Feedback sensors Pressure, temperature, position/limit switches and current sensing: location, units, range, calibration, rate and telemetry path.	Response / attachment ref:
F4 CONTROL	Mechanical interlocks Travel/pressure limits, overload protection, lockout conditions, safe sequencing and physical guarding.	Response / attachment ref:
F5 CONTROL	Loss behavior What happens on power loss, link loss, controller reboot and emergency stop; evidence that the implement reaches or remains in a safe state.	Response / attachment ref:
F6 FEATURE	Fault definitions Controller diagnostic codes, sensor-disconnect detection, manual reset and maintenance procedure.	Response / attachment ref:
F7 FEATURE	Other outputs Lights, horn, camera gimbal or additional actuators the app must display or control; identify protocol and safety constraints.	Response / attachment ref:

G Device, operations, software and evidence

This page defines the field operating constraints and the package to return with this form.

G1 NOW	Android hardware Phone/tablet model, Android version, screen size/orientation, mount, charging/power and planned connectivity at the operating site.	Response / attachment ref:
G2 FEATURE	Offline operation Locations/areas needing offline maps, map-data source/licence, expected internet outages, storage limits and update process.	Response / attachment ref:
G3 FEATURE	Mission workflow Who creates/approves routes and geofences; coordinate datum, field boundaries, no-go zones, transfer protocol and onboard execution limits.	Response / attachment ref:
G4 CONTROL	Identity and administration How operators and vehicles are provisioned, roles/permissions, device-loss procedure, signing/trust policy and key-rotation process. Do not send actual secrets.	Response / attachment ref:
G5 FEATURE	Diagnostics and records Required event/log retention, time synchronization, export format, service workflow and mandatory audit events.	Response / attachment ref:
G6 NOW	Evidence package Attach wiring/plumbing diagrams, controller datasheets, firmware/parameter exports, protocol references, redacted telemetry captures and labelled photos.	Response / attachment ref:
G7 NOW	Review ownership Name the hardware, firmware and safety reviewers; list unknowns, assumptions, planned tests and target date for verified answers.	Response / attachment ref: